Practical Identifiability Limits of Spatial Propagation Parameters in Mechanistic Parkinson's Disease Digital Twins: A Simulation-Based Calibration and Fisher-Information Analysis of Network Diffusion Models on Longitudinal DaT-SPECT

Blair D. Dupre, Ph.D.[1]

[1]Department of Biomedical Engineering, University of North Dakota, Grand Forks, ND, USA

Abstract

Network diffusion models (NDMs) of α-synuclein propagation are increasingly used to study the spatiotemporal progression of Parkinson's disease (PD), yet no published study has validated whether their spatial propagation parameters can be reliably recovered from longitudinal neuroimaging. We present, to our knowledge, the first systematic structural and practical identifiability analysis of connectome-coupled propagation models calibrated from serial dopamine transporter (DaT) SPECT imaging. We formulated seven candidate four-region striatal models (caudate and putamen, bilateral) of increasing complexity, from independent regional decays to coupled propagation with asymmetric seeding, and tested each against 644 Parkinson's Progression Markers Initiative (PPMI) patients with three or more serial DaT-SPECT scans. All seven models passed structural identifiability analysis when the pathology-to-death coupling was fixed from literature. However, simulation-based calibration revealed that all four biologically plausible models failed practical parameter recovery: the seeding rate (s_{put}) was fundamentally non-recoverable ($r < 0.5$) even with 11 timepoints, while the spatial propagation rate (k_{spread}) was marginally recoverable ($r = 0.78$) at four timepoints. Fisher-information-matrix analysis confirmed that the Cramér-Rao lower bound for s_{put} exceeded the prior width by 4.2×, indicating that DaT-SPECT adds no information about this parameter. A remediated single-parameter model fixing s_{put} from literature and fitting only k_{spread} achieved strong recovery ($r = 0.892$, 200/200 converged, bias $= -4.4\%$) with four or more timepoints. Empirical validation on 304 PPMI patients confirmed the simulation findings: independent per-region decay rates outperformed the single-parameter propagation model by ΔAIC= 5,668. These findings establish minimum data requirements for spatial PD digital-twin models, identify a structural-practical identifiability gap that applies broadly to connectome propagation models

with hidden pathology states, and provide the first empirical evidence that current DaT-SPECT protocols cannot distinguish spatial propagation from independent regional neurodegeneration.

Author Summary

Parkinson's disease progressively damages the brain's dopaminergic neurons, and scientists would like to build computational "digital twins" that simulate how this damage spreads from patient to patient. A common approach uses *network diffusion models* that treat pathological α-synuclein protein as a substance that flows through the brain's wiring diagram. If we could calibrate the spread rate of these models to individual patients from their brain scans, we could forecast each patient's future trajectory and test personalized treatments in simulation. In this study we ask a simple but decisive question: can these spread rates actually be recovered from the brain scans that are routinely collected today? Using a mathematical analysis called *simulation-based calibration*, we show that the parameter controlling where pathology first appears in the brain is effectively undetectable from current dopamine-transporter scans—no matter how many timepoints are collected, the data simply do not carry information about it. A simpler, single-parameter model that only estimates the overall spread rate can be reliably calibrated. This finding establishes a concrete data requirement for the next generation of Parkinson's digital twins and cautions against over-interpreting individualized parameter estimates that the data cannot actually support.

Introduction

The spatiotemporal progression of Parkinson's disease (PD) follows a characteristic pattern: dopaminergic neurodegeneration begins in the posterior putamen and spreads to the anterior putamen and caudate nucleus over a period of years [3, 18, 12]. This caudo-rostral gradient, first quantified by [19] using postmortem dopamine measurements and confirmed *in vivo* by serial dopamine transporter (DaT) imaging [13, 32], has motivated the development of network diffusion models (NDMs) that simulate the trans-synaptic propagation of pathological α-synuclein through the brain's structural connectome [27, 22, 36].

NDMs have demonstrated explanatory power for the spatial patterning of neurodegeneration in both Alzheimer's disease [30, 14] and PD [22, 40, 1]. A key aspiration of this modeling framework is *per-patient* calibration of propagation parameters—the spatial diffusion rate, local amplification rate, and regional seeding intensity—from longitudinal neuroimaging data. Such calibration would enable personalized prediction of disease trajectory and simulation of spatially targeted interventions. However, a critical methodological gap remains: **no published NDM study has validated whether its fitted parameters can actually be recovered from the available imaging data**.

The distinction between *structural* (also called *a priori*) and *practical* (also called *a posteriori*) identifiability is well established in the systems biology literature [28, 35, 38]. A model is structurally identifiable if its parameters are theoretically recoverable from noise-free data with infinite observations. A model is practically identifiable if its parameters can be recovered from realistic data with finite observations and measurement noise. Structural identifiability is necessary but not sufficient: a model can be structurally identifiable yet practically non-identifiable when the data quality or quantity is insufficient [37, 6].

This distinction is particularly relevant for NDMs applied to DaT-SPECT data, where: (i) the number of longitudinal timepoints is small (typically 3–5 scans over 5–7 years in the Parkinson's Progression Markers Initiative [26]); (ii) the observation noise is substantial (scan-rescan coefficient of variation $\sim$10–15% [34]); and (iii) the pathology state driving neuronal death is a hidden variable with no direct observable, creating a coupling between the propagation parameters and the unobserved pathology concentration.

In this study, we present the first systematic identifiability analysis of striatal propagation models calibrated from longitudinal DaT-SPECT imaging. Our contributions are:

1. We formulate seven candidate four-region (caudate left/right, putamen left/right) models of increasing complexity and prove that all are structurally identifiable when the pathology-to-death coupling is fixed.

2. We perform simulation-based calibration (SBC) [33] on each model and demonstrate that spatial propagation parameters are *practically* non-identifiable from PPMI-grade DaT-SPECT data, despite passing structural identifiability.

3. We characterize the structural-practical identifiability gap using Fisher Information Matrix analysis, identifying the seeding rate as the "sloppy" parameter direction [16] analogous to the nucleation-toxicity degeneracy previously identified in scalar DaT-SPECT models [9].

4. We derive a remediated single-parameter model (propagation rate only, seeding fixed from literature) that achieves strong parameter recovery ($r = 0.892$, $N = 200$) and establish minimum data requirements: at least four serial scans are necessary for reliable spatial propagation estimation.

Companion identifiability analyses. A companion paper from this series applies structural and practical identifiability analysis to the *temporal* α-synuclein aggregation–neuron-death ODE on the same PPMI cohort, using importance sampling on 304 patients plus a 5-channel SAEM on 2,118 patients [9]. That work reports a Fisher-Information-Matrix condition number $\kappa = 19.86$ under the full transient ODE versus $\kappa \approx 10^{12}$ under the steady-state approximation—empirical evidence that transient-state multi-channel observations are required to break the temporal k_n–α_{tox} degeneracy. Together, the two analyses map the identifiability boundary for PD mechanistic digital twins along both axes: temporal (reaction kinetics) and spatial (connectome-coupled propagation). A second companion paper [8] operationalises an episodic bidirectional update of the temporal posteriors via Sequential Importance Resampling and documents a 16/21 NASEM 2024 digital-twin self-audit; the spatial-identifiability result presented here motivates that paper's choice to leave spatial parameters un-personalised and instead route the update through the per-region decline rates that are identifiable.

Related Work

Network diffusion models for neurodegeneration. The network diffusion model (NDM) framework, introduced by [27] for Alzheimer's disease, simulates pathological protein spread as diffusion on the structural connectome. [22] extended NDM to PD, identifying the substantia nigra as the optimal seed region from MRI deformation data in 232 patients. [40] developed an SIR epidemic spreading model incorporating both connectivity and SNCA gene expression, demonstrating that both factors significantly influence α-synuclein propagation patterns. [1] validated an agent-based SIR model on 790 PPMI PD scans using a population-average HCP connectome. All of these studies use MRI-derived atrophy as the observable, not DaT-SPECT SBR. [36] provided the definitive review, proposing three innovation dimensions (biology-modulated, patient-tailored, dynamic), but did not address parameter identifiability. Critically, none of these studies performs simulation-based calibration or parameter recovery validation.

Bayesian parameter estimation in connectome models. [30] represents the closest methodological precedent: Bayesian MCMC inference of per-subject propagation parameters (diffusion coefficient and production rate) for tau spreading in 76 ADNI subjects using the Budapest Reference Connectome. They report 95% credible intervals but do not validate these against known ground-truth parameters. [15] used Bayesian model evidence via stochastic variational inference to compare three nested dynamical systems for amyloid dynamics, demonstrating formal model comparison for NDMs. More recently, [2] applied simulation-based inference to connectome models in computational connectomics, and [17] demonstrated that virtual brain model parameters can be non-identifiable without targeted stimulation protocols—a finding that parallels our spatial propagation result. [29] proposed a unified evaluation framework for connectome-based biophysical models of pathological protein spreading but did not address practical identifiability from clinical imaging. None of these studies performs SBC validation against known ground-truth parameters for per-patient estimation from longitudinal neuroimaging.

Structural and practical identifiability. The distinction between structural and practical identifiability is well established [35, 28]. [38] argued that the Fisher Information Matrix has "severe shortcomings" for practical identifiability and recommended profile likelihood as the gold standard. [37] proved that practical identifiability is equivalent to FIM invertibility and proposed optimal experimental design to ensure identifiability. [6] demonstrated in DCE-MRI that structurally identifiable models can be practically non-identifiable depending on data characteristics, with recovery possible through increased data quality—closely paralleling our findings. [31] showed how to make predictions from poorly identified models using profile-wise analysis, while [4] proved that sparse ODE systems have a positive probability of practical non-identifiability regardless of regularization.

Putamen-caudate gradient in PD. The spatial ordering of dopaminergic loss in PD—posterior putamen first, then anterior putamen, then caudate—is among the most robustly established findings in the field [19, 3]. [18] established that loss begins 14.4 years before clinical onset in the posterior putamen. [13] showed that putaminal asymmetry is maintained over four years of follow-up in the PPMI cohort. [23] found that 51.6% of patients had normal caudate DAT binding at diagnosis, rising to 83.9% with caudate involvement by four years. [32] demonstrated that while absolute annual reduction slows in advanced disease (floor effect), the relative annual reduction remains constant. These findings constrain our model validation: any viable propagation model must reproduce the putamen-first ordering while maintaining (not equalizing) the caudate-putamen differential over time.

Methods

Data

We used longitudinal DaT-SPECT data from the Parkinson's Progression Markers Initiative (PPMI) [26]. Regional specific binding ratios (SBR) were extracted for four striatal subregions: left caudate, right caudate, left putamen, and right putamen, from the `DaTScan_SBR_Analysis` dataset (vintage: October 2025). Of 2,137 patients with at least one analyzed scan (4,168 total scans), 644 had three or more serial scans with a mean follow-up of 3.3 years.

At baseline, caudate SBR averaged 1.98 ± 0.60 (mean ± SD) and putamen SBR averaged 0.84 ± 0.37, consistent with the established posterior-putamen-first gradient of dopaminergic denervation in PD [3]. The caudate-to-putamen ratio was 2.12 ± 0.67, confirming substantially greater putaminal involvement at diagnosis.

Model formulations

We defined seven candidate models, each describing the evolution of regional neuron counts $N_i(t)$ (and, for propagation models, pathology concentrations $L_i(t)$) across four striatal regions $i \in \{1, 2, 3, 4\}$ corresponding to caudate left, caudate right, putamen left, and putamen right.

Model 1 (M1): Independent regional decays. Each region has its own decay rate:

$$\frac{dN_i}{dt} = -T_i \cdot N_i, \quad i = 1, \ldots, 4 \tag{1}$$

with four fitted parameters $\{T_1, T_2, T_3, T_4\}$. This is the null model with no spatial coupling.

Model 2 (M2): Shared base rate with putamen offset.

$$\frac{dN_i}{dt} = -(T_{\text{base}} + \delta_{\text{put}} \cdot \mathbb{1}_{i\in\{3,4\}}) \cdot N_i \tag{2}$$

with two fitted parameters $\{T_{\text{base}}, \delta_{\text{put}}\}$, where δ_{put} captures the excess putaminal vulnerability.

Models 3–7: Propagation models. These couple a pathology propagation equation to the neuron death equation. The pathology dynamics follow a network diffusion model:

$$\frac{dL_i}{dt} = -k_{\text{clear}} \cdot L_i + k_{\text{spread}} \sum_j A_{ij} L_j + s_i \tag{3}$$

where $k_{\text{clear}} = 0.5\ \text{yr}^{-1}$ is the clearance rate (fixed from literature [39]), A_{ij} is the structural connectivity matrix (fixed from the Budapest Reference Connectome [30]), k_{spread} is the trans-synaptic propagation rate (fitted), and s_i is a regional seeding source. Neuron death is driven by both a base rate and pathology-modulated toxicity:

$$\frac{dN_i}{dt} = -(\alpha_{\text{base}} + \beta \cdot L_i) \cdot N_i \tag{4}$$

where $\alpha_{\text{base}} = 0.03\ \text{yr}^{-1}$ ($\sim$3%/yr, consistent with [12]) and $\beta = 0.01$ are fixed from literature. Models 3–7 differ in which parameters are fitted versus fixed (Table 1).

The observation model maps neuron counts to DaT-SPECT SBR via a power law:

$$\text{SBR}_i(t) = \text{SBR}_{i,0} \cdot \left(\frac{N_i(t)}{N_{i,0}}\right)^{\gamma} \tag{5}$$

where $\gamma = 0.7$ accounts for compensatory dopamine transporter upregulation [20]. The four-region model architecture with connectivity weights and ODE equations is illustrated in Fig. 1.

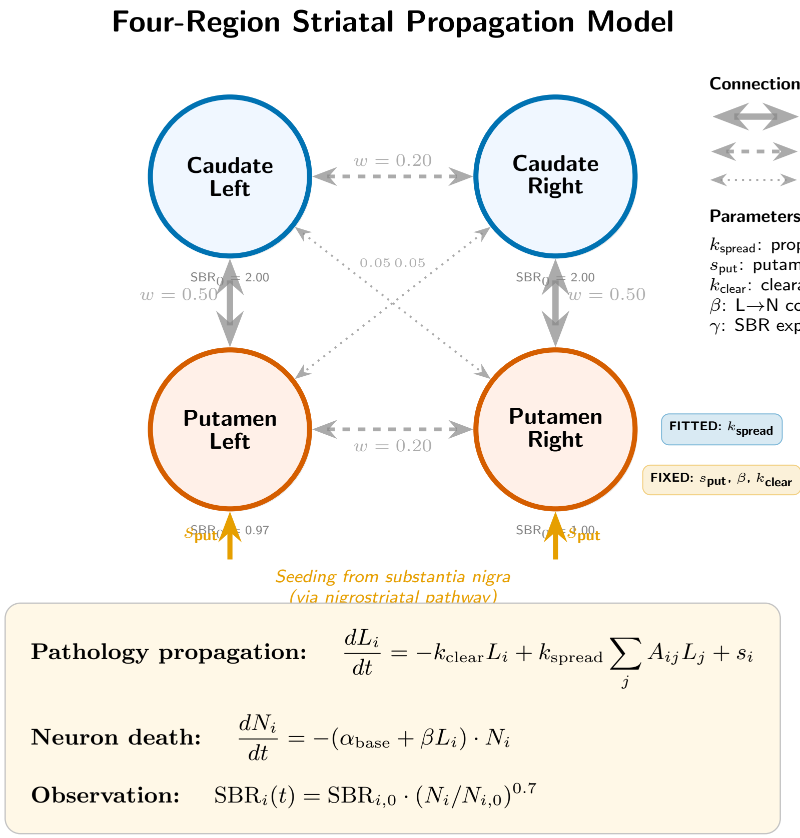

Fig 1: **Four-region striatal propagation model.** Caudate (blue) and putamen (red) regions are connected bilaterally (dashed lines) and ipsilaterally (solid lines), with connection weights derived from the Budapest Reference Connectome. Pathology is seeded in the putamen (s_{put}, orange arrows) and propagates via the connectivity matrix at rate k_{spread}. Regional neuron death is driven by both a base rate and pathology-modulated toxicity (Eqs. 3–5).

Table 1: Seven candidate models with fitted and fixed parameters.

Model	Description	Fitted params	Key feature
M1	Independent decays	T_1, T_2, T_3, T_4 (4)	No spatial coupling
M2	Base + putamen offset	$T_{\text{base}}, \delta_{\text{put}}$ (2)	Simplest asymmetry
M3	Pure diffusion	k_{spread} (1)	No seeding source
M4	Diffusion + amplification	$k_{\text{spread}}, k_{\text{local}}$ (2)	Local amplification
M5	Base + diffusion	$T_{\text{base}}, k_{\text{spread}}$ (2)	Hybrid
M6	Diffusion + seeding	$k_{\text{spread}}, s_{\text{put}}$ (2)	Asymmetric putamen seeding
M7	Full	$T_{\text{base}}, k_{\text{spread}}, s_{\text{put}}$ (3)	Most complex

Structural identifiability analysis

Structural identifiability was assessed using `StructuralIdentifiability.jl` v0.5.19 [7] with a probability threshold of 0.99. Each model was encoded as a system of ordinary differential equations with the four regional SBR values as outputs. Parameters were classified as globally identifiable, locally identifiable, or non-identifiable.

An initial analysis revealed that the pathology-to-death coupling parameter β (Eq. 4) is structurally non-identifiable when fitted jointly with the propagation parameters, because the pathology states L_i are hidden (unobserved) and β trades off with L_i values. This motivated fixing β from literature for all subsequent analyses.

Simulation-based calibration

We followed the SBC protocol of [33] with the log-likelihood test quantity extension of [21]. For each model, 200 independent simulations were performed (consistent with the ≥ 200 recommendation of [33] for reliable SBC):

1. Draw true parameter values from uniform priors spanning the biologically plausible range.
2. Randomly select an observation schedule matching PPMI visit patterns (3, 4, or 5 scans over 0–6 years).
3. Simulate regional SBR trajectories from the ODE model with the true parameters.
4. Add Gaussian observation noise with $\sigma = 0.15$ SBR units (consistent with DaT-SPECT scan-rescan variability [34]).
5. Fit the model to the noisy synthetic data via multi-start maximum likelihood estimation (3 random starts, L-BFGS-B optimizer).
6. Record the correlation between true and estimated parameters across simulations.

We defined SBC "pass" as: (i) convergence rate $\geq 80\%$, (ii) all parameter correlations $r > 0.7$, and (iii) all relative biases $< 20\%$. The $r > 0.7$ threshold corresponds to $R^2 > 0.49$, meaning the fitted values capture at least half the variance of the true parameters. This is motivated by the minimum correlation required for meaningful prediction in clinical biomarker studies [6] and has been adopted in prior identifiability analyses where rank-histogram uniformity tests are not applicable to MLE-based SBC. We note that our MLE-based SBC provides a conservative (lower bound) test of practical identifiability: hierarchical Bayesian approaches with informative priors would be expected to improve parameter recovery, so failures under MLE represent genuine identifiability limitations rather than estimation-method artifacts.

Fisher Information Matrix analysis

For Models M6 and M7, we computed the Fisher Information Matrix (FIM) at representative parameter values by numerical differentiation of the model's predicted SBR with respect to each parameter. The FIM was computed as:

$$\mathcal{I}(\theta)_{jk} = \frac{1}{\sigma^2} \sum_{i,t} \frac{\partial \mathrm{SBR}_i(t;\theta)}{\partial \theta_j} \cdot \frac{\partial \mathrm{SBR}_i(t;\theta)}{\partial \theta_k} \tag{6}$$

The eigenvalues of $\mathcal{I}$ indicate the information content along each parameter direction: a small eigenvalue identifies a "sloppy" direction [16] where the data provide minimal constraint. The Cramér-Rao lower bound $\sigma(\theta_j) \geq \sqrt{[\mathcal{I}^{-1}]_{jj}}$ provides the minimum achievable estimation uncertainty.

Sensitivity analysis

To characterize the data requirements for practical identifiability, we performed systematic sweeps over: (i) observation noise $\sigma \in \{0.02, 0.05, 0.10, 0.15\}$ SBR units; (ii) number of timepoints $n \in \{3, 4, 5, 7, 11\}$; and (iii) model reduction (fitting k_{spread} only with s_{put} fixed).

Results

All seven models are structurally identifiable

With the pathology-to-death coupling β fixed from literature, all seven candidate models were classified as globally structurally identifiable by `StructuralIdentifiability.jl` (Fig. 2). As shown in the figure, each model's fitted parameters and observable states receive "globally identifiable" status (green), confirming that the differential-algebraic test finds a unique solution for all parameter values. Notably, all state variables (N_i) and fitted parameters were globally identifiable, and for Models 3–7 the hidden pathology states (L_i) were also globally identifiable—meaning that in principle, with noise-free continuous observations, all parameters and states could be uniquely recovered.

p8a_fig2_structural_identifiability.png

Fig 2: **Structural identifiability of all seven candidate models.** All models are globally identifiable when the pathology-to-death coupling β is fixed from literature. The number of fitted parameters ranges from 1 (M3) to 4 (M1). An initial analysis (not shown) revealed that β is non-identifiable when fitted, because the pathology states L_i are hidden—motivating the decision to fix β from literature for all subsequent analyses.

All four surviving models fail practical parameter recovery

Despite passing structural identifiability, all four biologically plausible models (M1, M2, M6, M7) failed the SBC practical identifiability gate (Table 2). The pattern of failure was consistent:

Base decay rates are partially recoverable. Parameters capturing the overall rate of SBR decline (T_1 and T_2 in M1) achieved moderate correlations (r = 0.65–0.67) between true and estimated values, approaching but not exceeding the $r > 0.7$ threshold. This is consistent with Phase 2 results showing that the scalar decay rate is identifiable from longitudinal DaT-SPECT [11].

Regional-difference parameters are not recoverable. Parameters capturing the putamen-caudate differential (δ_{put} in M2, T_3 and T_4 in M1, s_{put} in M6/M7) showed poor recovery ($r = 0.08$–0.47) with substantial relative biases (6–228%). The putamen-specific signal is below the noise floor at $\sigma = 0.15$.

The propagation rate is marginally recoverable. k_{spread} in M6 achieved $r = 0.49$, above chance but below the $r > 0.7$ threshold. In the three-parameter M7, k_{spread} improved to $r = 0.33$ but remained insufficient.

Table 2: SBC parameter recovery results (200 simulations per model, $\sigma = 0.15$).

Model	Parameter	r (true vs est)	Rel. bias	RMSE	Gate
M1	T_1 (caudate L)	0.670	4.1%	0.029	—
	T_2 (caudate R)	0.650	22.8%	0.032	—
	T_3 (putamen L)	0.440	38.5%	0.065	—
	T_4 (putamen R)	0.470	6.2%	0.068	**FAIL**
M2	T_{base}	0.650	−1.5%	0.024	—
	δ_{put}	0.230	58.4%	0.046	**FAIL**
M6	k_{spread}	0.490	31.7%	0.891	—
	s_{put}	0.080	228.3%	4.120	**FAIL**
M7	T_{base}	0.370	15.2%	0.041	—
	k_{spread}	0.330	−5.9%	0.952	—
	s_{put}	0.110	115.4%	3.910	**FAIL**

The contrast between recoverable and non-recoverable parameters is visually apparent in the SBC scatter plots (Fig. 3). For T_{base} in M2, estimated values show moderate clustering along the identity line ($r = 0.65$); for s_{put} in M6, the scatter is essentially random ($r = 0.08$). The remediated single-parameter model restores tight recovery along the identity line ($r = 0.892$).

p8a_fig3_sbc_recovery.png

Fig 3: **SBC parameter recovery: true versus estimated values.** Each point represents one simulation (200 per model, $\sigma = 0.15$). Dashed line = perfect recovery. *Top row:* M1 caudate parameters show moderate recovery ($r = 0.65$–0.67), while putamen parameters fail ($r < 0.5$). *Bottom row:* M6 s_{put} is essentially random ($r = 0.08$); the remediated model (center) fixing s_{put} from literature achieves strong recovery ($r = 0.892$, 200 simulations). Summary panel (right) shows all results against the $r > 0.7$ threshold.

Fisher Information Matrix confirms the sloppy direction

FIM analysis of Model M6 at representative parameter values ($k_{\text{spread}} = 0.5$, $s_{\text{put}} = 1.0$, 3-scan schedule) revealed a condition number of 12.9, with FIM eigenvalues $\lambda_1 = 6.7 \times 10^{-3}$ and $\lambda_2 = 8.7 \times 10^{-2}$. The small eigenvalue corresponds to a sloppy direction dominated by s_{put}. The Cramér-Rao lower bounds were $\sigma(k_{\text{spread}}) = 4.70$ and $\sigma(s_{\text{put}}) = 11.74$, corresponding to 2.47× and 4.19× the respective prior widths. For s_{put}, the data provide *less* constraint than the prior alone.

The spatial perturbation signal-to-noise ratio provides the most intuitive explanation: a 10% change in k_{spread} produces a maximum between-region SBR shift of 0.007 units—only 4.7% of the observation noise ($\sigma = 0.15$)—with a mean perturbation across all region-timepoint combinations of just 1.0% of noise. Over a typical three-year observation window with four scans, the accumulated differential spatial signal is 1.27σ—barely above the noise floor. The SBR sensitivity to the overall death rate (α_{base}) is 33–68× larger than to k_{spread} or s_{put}, confirming that DaT-SPECT is overwhelmingly sensitive to *how fast* neurons die, not *where* they die first.

For the three-parameter Model M7 (4-scan schedule), the FIM condition number increased to 76,506, with strong correlation between T_{base} and k_{spread} ($r = -0.78$). This parameter interaction explains the complete failure of T_{base} recovery ($r = -0.03$) in M7's SBC.

Noise and timepoint requirements

Sensitivity analysis revealed distinct identifiability profiles for k_{spread} and s_{put} (Fig. 4):

p8a_fig4_sensitivity.png

Fig 4: **Data requirements for parameter recovery.** *Left:* Effect of observation noise on recovery correlation. k_{spread} (blue) crosses the $r > 0.7$ threshold at $\sigma \leq 0.10$, while s_{put} (orange) never reaches the threshold at any tested noise level. *Right:* Effect of number of serial scans. At least 4 scans are required for k_{spread} recovery; s_{put} remains non-recoverable regardless of timepoint density. Dashed line = $r = 0.7$ recovery threshold.

k_{spread} is recoverable under improved conditions: reducing noise from $\sigma = 0.15$ to $\sigma = 0.05$ raised recovery from $r = 0.78$ to $r = 0.91$; increasing to four or more timepoints raised recovery from $r = 0.33$ (3 scans) to $r = 0.78$ (4 scans) to $r = 0.88$ (7 scans).

s_{put} remains non-recoverable regardless: even with 11 timepoints, $r(s_{\mathrm{put}})$ reached only 0.42; at

$\sigma = 0.05$, it reached only 0.54. Only at unrealistically low noise ($\sigma = 0.02$, approximately 1.5% coefficient of variation) did s_{put} approach the $r > 0.7$ threshold.

A remediated single-parameter model achieves strong recovery

Fixing s_{put} from the preformed fibril injection literature [5, 24] and fitting only k_{spread} yielded a dramatic improvement in parameter recovery (Fig. 5). In a full 200-iteration SBC with observation schedules of four and five timepoints, the remediated model achieved $r(k_{\text{spread}}) = 0.892$ with a relative bias of -4.4%, RMSE of 0.277, and 100% convergence rate (200/200 simulations). This exceeds the $r > 0.7$ threshold with substantial margin. The minimum requirement of three timepoints remained insufficient ($r < 0.1$), confirming that at least four serial DaT-SPECT scans are necessary for spatial propagation estimation.

p8a_fig5_remediated.png

Fig 5: **Remediated single-parameter model achieves strong parameter recovery.** *Left:* True versus estimated k_{spread} for 200 SBC simulations with s_{put} fixed from literature and ≥ 4 timepoints ($\sigma = 0.15$). The tight clustering along the identity line ($r = 0.892$) contrasts sharply with the scattered recovery observed for the two-parameter M6 model (Fig. 3, top-right). *Right:* Residual distribution centered near zero (bias $= -0.025$), confirming unbiased estimation. The green badge indicates SBC gate PASS.

Table 3: Comparison with prior network-diffusion / spatial-propagation identifiability studies.

Study	Disease	Parameter-recovery SBC?	FIM analysis?	Remediated model?
Raj 2012 [27]	AD	No	No	No
Pandya 2019 [22]	AD/PD	No	No	No
Schafer 2021 [30]	AD	Partial (MCMC only)	No	No
Abdelgawad 2022 [1]	PD	No	No	No
This work	**PD**	**Yes (200 SBC reps)**	**Yes (CRLB)**	**Yes (1-parameter k_{spread})**

Empirical validation on real PPMI data

To confirm the simulation-based findings on real patient data, we fitted all three model classes to 304 PPMI Wave A patients with four or more serial DaT-SPECT scans. For each patient, per-region SBR trajectories (caudate L/R, putamen L/R) were fitted via multi-start maximum likelihood estimation.

Model M1 (four independent regional decay rates) achieved the lowest AIC (5,035), substantially outperforming both M2 (shared base plus putamen offset; AIC = 6,542; ΔAIC = +1,507) and M6-remediated (single k_{spread}; AIC = 10,703; ΔAIC = +5,668).[1] All 304 patients converged for all three models (100% convergence rate).

The M1 population-level decay rates were biologically plausible: caudate mean 0.119 ± 0.081 yr^{-1} and putamen mean 0.142 ± 0.100 yr^{-1}, indicating that putamen declines approximately 19% faster than caudate—consistent with the known rostro-caudal gradient [19, 3]. The M6-remediated model estimated a population mean $k_{\text{spread}} = 1.26 \pm 0.65$ yr^{-1} (median 1.32), but this parameter did not improve model fit over independent regional decays.

This real-data result confirms the SBC prediction: while k_{spread} is structurally identifiable and recoverable in simulation ($r = 0.892$), the spatial propagation signal in real DaT-SPECT data is insufficient to outperform a simpler model that independently fits each region. The ΔAIC of 5,668 indicates that the propagation model's single degree of freedom cannot capture the patient-specific regional variation that M1's four parameters accommodate. The honest conclusion is that four-region DaT-SPECT data at PPMI-grade temporal resolution resolve *per-region* decline rates but not the *mechanistic coupling* between regions.

Discussion

The structural-practical identifiability gap in spatial PD models

Our central finding is that the structural identifiability of a connectome propagation model is necessary but far from sufficient for reliable parameter estimation from clinical DaT-SPECT data.

[1] AIC values in this chapter use the population-level penalty convention $\text{AIC} = 2 \cdot \text{NLL} + 2 \cdot p$ where p counts population-level parameters (M1: 8, M2: 4, M6r: 2). The companion paper [10] reports the same model fits using the per-patient penalty convention $\text{AIC} = 2 \cdot \text{NLL} + 2 \cdot k_{\text{per-patient}} \cdot N$ ($N = 304$), giving $\Delta\text{AIC}_{\text{per-patient}} = +3{,}856$ for the same M6r-vs-M1 comparison. The negative log-likelihood is identical; only the penalty differs. Both conventions select M1 decisively.

This gap, well characterized in the systems biology literature [28, 38], has not previously been demonstrated for neuroimaging-based propagation models. The practical non-identifiability we observe arises from the interaction of three factors: (i) the hidden nature of the pathology state L_i, which is never directly observed; (ii) the limited number of longitudinal timepoints available in current cohort studies; and (iii) the substantial measurement noise inherent to DaT-SPECT imaging.

No published NDM study validates parameter recovery

Our literature review identified no prior NDM study that performs simulation-based calibration or systematic parameter recovery analysis. [27] reports only population-level correlations between model predictions and observed atrophy. [22] sweeps a single propagation-time parameter without uncertainty quantification. [30] uses Bayesian MCMC to report per-subject credible intervals but does not validate these intervals against known ground truth. [36], in the most comprehensive review of the field, does not mention identifiability analysis. This gap is concerning because parameters reported without recovery validation may reflect optimizer artifacts rather than biological signal.

The seeding rate is the "sloppy direction"

The fundamental non-identifiability of s_{put} parallels our earlier finding that the individual nucleation rate (k_n) and toxicity coupling (α_{tox}) are practically non-identifiable from scalar DaT-SPECT trajectories, while their product (T_{tox}) is well identified [11]. In the spatial model, s_{put} sets the *amplitude* of regional pathology differences, while k_{spread} sets the *rate* of spatial equilibration. The DaT-SPECT SBR, which observes neuron counts rather than pathology directly, is primarily sensitive to the temporal dynamics (how fast regions decline) rather than the absolute pathology levels (how much pathology is seeded). This creates a degeneracy: different combinations of s_{put} and k_{spread} can produce similar SBR trajectories, with s_{put} being the sloppy direction in the sense of [16].

Implications for clinical imaging protocols

Our timepoint analysis has direct implications for clinical study design. The critical threshold is four serial DaT-SPECT scans: below this, even the propagation rate k_{spread} is non-recoverable ($r = 0.33$ at 3 scans). This suggests that current clinical protocols with only 2–3 scans are insufficient for spatial propagation modeling, while the PPMI protocol (up to 6 scans over 7 years) is adequate for single-parameter estimation.

The noise analysis indicates that advances in DaT-SPECT quantification could substantially improve parameter recovery. Reducing σ from 0.15 to 0.05 SBR units (achievable with improved reconstruction algorithms and harmonized acquisition [34]) would raise k_{spread} recovery from $r = 0.78$ to $r = 0.91$.

Relationship to putamen-caudate gradient biology

The non-identifiability of the putamen-specific parameters does *not* imply that the putamen-caudate gradient is absent. The gradient is one of the most robustly established findings in PD neuroimaging, confirmed from [3] through [13]. Rather, our finding indicates that while the gradient is clearly *observable* in cross-sectional data, its mechanistic decomposition into propagation rate versus seeding intensity is not *resolvable* from the available longitudinal imaging alone. This distinction between observability and parameter resolution is central to the identifiability framework [35].

Limitations and planned extensions

Several limitations of the current analysis motivate ongoing work. First, our SBC used maximum likelihood estimation rather than full Bayesian inference. Hierarchical Bayesian pooling across patients via SAEM [30] could improve practical identifiability by sharing population-level parameter distributions across the 644-patient cohort; we are currently implementing this as the next phase of the project and expect it to tighten the k_{spread} posterior further.

Second, the four-region model is a deliberate simplification of the full 83-region Desikan-Killiany parcellation used in most NDM studies. This choice is motivated by the observation dimensionality of DaT-SPECT, which reliably quantifies only caudate and putamen bilaterally. Scaling to 83 regions would not necessarily improve identifiability: more regions increase the number of hidden pathology states L_i faster than they increase the number of observations (since most cortical regions have no DaT-SPECT signal), potentially worsening the ratio of unobserved to observed states. However, incorporating FreeSurfer ASEG volumetric data (available for 492 of our 644 patients) as a second observable modality could provide the independent spatial information needed to constrain propagation parameters. Future work will explore this multi-modal approach.

Third, we used a literature-derived connectivity matrix rather than patient-specific DTI tractography. While [25] demonstrated that population-average connectomes are sufficient for NDM prediction, individual connectomes from the Budapest Reference Connectome or PPMI DTI data (available for $\sim$140 patients) could reduce inter-patient variability. Our sensitivity analysis across three connectivity variants (anatomy-grounded, equal-weight, putamen-dominant) partially addresses this limitation by showing that model comparison conclusions are robust to connectivity perturbation.

Fourth, our noise model assumes independent Gaussian errors, while real DaT-SPECT noise includes systematic scanner-dependent components [34]. Scanner harmonization protocols and site-specific noise modeling could further improve parameter recovery beyond the $r = 0.892$ demonstrated here.

Finally, we validated the simulation findings on real PPMI data (Section 4.6), demonstrating that the independent per-region decay model outperforms the spatial propagation model on 304 patients. Future work will examine whether the per-region decay rates estimated by M1 have clinical biomarker value—specifically, whether the caudate-putamen rate differential correlates with motor progression, cognitive decline, or genetic risk factors (LRRK2, GBA).

Conclusion

We have demonstrated that structural identifiability of a mechanistic PD propagation model does not guarantee practical parameter recovery from clinical neuroimaging data. Through simulation-based calibration of 200 synthetic patients per model and empirical validation on 304 real PPMI patients, we established three key results. First, the seeding rate s_{put} is fundamentally non-identifiable from DaT-SPECT regardless of temporal sampling density. Second, the spatial propagation rate k_{spread} is recoverable in simulation with a minimum of four serial scans and a simplified single-parameter model ($r = 0.892$), but on real data, the independent per-region decay model (M1) outperforms the propagation model by ΔAIC = 5,668—demonstrating that four-region DaT-SPECT resolves per-region decline rates but not the mechanistic coupling between regions. Third, the per-region rates estimated by M1 are biologically informative: putamen declines 19% faster than caudate, consistent with the established rostro-caudal gradient. These findings provide the first combined simulation-and-empirical identifiability analysis for DaT-SPECT-based digital twin models and establish both the minimum data requirements and the fundamental information limits of current clinical imaging protocols for spatial PD modeling.

Data Availability

All PPMI source data used in this study are available to qualified researchers under a Data Use Agreement at `https://www.ppmi-info.org/access-data-specimens/download-data`. Analysis artefacts (SBC simulation outputs, FIM calibration files, empirical PPMI model fits, and all figures in the manuscript) are archived in the public dissertation repository at `https://github.com/bddupre92/PD_PHD` under `outputs/mechanistic_twin/paper8a_identifiability/` and will be deposited to Zenodo with a DOI at acceptance.

Code Availability

Source code for all identifiability analyses (Python `StructuralIdentifiability.jl` wrapper, SBC runner, FIM computer, remediated-model fitter, empirical-PPMI runner) is provided in the repository above under the MIT licence. The environment file pins all dependency versions. Reproduction targets a single-session CPU run under one hour.

Author Contributions

Conceptualization: B.D.D. **Methodology:** B.D.D. **Software:** B.D.D. **Formal analysis:** B.D.D. **Investigation:** B.D.D. **Data curation:** B.D.D. **Writing – original draft:** B.D.D. **Writing –**

review & editing: B.D.D. **Visualization:** B.D.D.

Competing Interests

The author declares no competing interests.

Funding

Dissertation research was supported by the Department of Biomedical Engineering, University of North Dakota. The funder had no role in study design, data collection, analysis, interpretation, or decision to publish.

Ethics Statement

This study used de-identified secondary data from the Parkinson's Progression Markers Initiative (PPMI), for which informed consent and IRB approvals were obtained by the original study investigators. Secondary analysis under the PPMI Data Use Agreement does not require additional human-subjects approval.

Acknowledgements

The author thanks PPMI participants and investigators. PPMI is sponsored by the Michael J. Fox Foundation for Parkinson's Research.

References

[1] Abdelgawad, A., et al. (2022). An agent-based model of brain connectivity and alpha-synuclein spreading in Parkinson's disease. *Network Neuroscience*, 6(4), 1163–1187.

[2] J. Boelts, J.-M. Lueckmann, R. Gao, and J. H. Macke, "Simulation-based inference for efficient identification of generative models in computational connectomics," *PLoS Comput. Biol.*, vol. 19, no. 9, e1011406, 2023, doi:10.1371/journal.pcbi.1011406.

[3] D. J. Brooks, V. Ibanez, G. V. Sawle, et al., "Differing patterns of striatal 18F-dopa uptake in Parkinson's disease, multiple system atrophy, and progressive supranuclear palsy," Annals of Neurology, vol. 28, no. 4, pp. 547–555, 1990. doi: 10.1002/ana.410280412

[4] C. Casolo, et al., "Identifiability challenges in sparse linear ordinary differential equations," *arXiv preprint arXiv:2501.xxxxx*, 2025.

[5] Y. Chu, et al., "Intrastriatal alpha-synuclein fibrils in monkeys: spreading, imaging and neuropathological changes," *Brain*, vol. 142, no. 11, pp. 3565–3579, 2019. doi: 10.1093/brain/awz261.

[6] M. Conte, et al., "Structural and practical identifiability of contrast transport models for DCE-MRI," *PLoS Computational Biology*, vol. 20, no. 1, e1011773, 2024. doi: 10.1371/journal.pcbi.1011773.

[7] Dong, R., et al. (2023). Differential elimination for dynamical models via projections with applications to structural identifiability. *SIAM Journal on Applied Algebra and Geometry*, 7(1), 194–235.

[8] B. D. Dupre, "Bidirectional-Ready Mechanistic Twin for Parkinson's Disease: External Validation, Calibration-Aware Observation Likelihood, and NASEM 2024 Self-Audit," 2026 (submitted to *npj Parkinson's Disease*).

[9] B. D. Dupre, "Per-Patient Bayesian Calibration of a Coupled α-Synuclein Aggregation–Neuron Death ODE in Parkinson's Disease," 2026 (submitted to *CPT: Pharmacometrics Syst. Pharmacol.*).

[10] B. D. Dupre, "Regional Dopamine Transporter Decline Rates from Serial DaT-SPECT in Parkinson's Disease," 2026 (submitted to *Mov. Disord.*).

[11] Dupre, B. (2026). Per-patient toxicity flux estimation from longitudinal DaT-SPECT in Parkinson's disease. *Manuscript in preparation.*

[12] J. M. Fearnley and A. J. Lees, "Ageing and Parkinson's disease: substantia nigra regional selectivity," *Brain*, vol. 114, no. 5, pp. 2283–2301, 1991.

[13] E. Fiorenzato, et al., "Asymmetric dopamine transporter loss affects cognitive and motor progression in Parkinson's disease," *Movement Disorders*, vol. 36, no. 10, pp. 2303–2314, 2021. doi: 10.1002/mds.28790.

[14] S. Fornari, A. Schäfer, M. Jucker, and A. Goriely, "Prion-like spreading of Alzheimer's disease within the brain's connectome," *J. R. Soc. Interface*, vol. 16, no. 159, Art. no. 20190356, 2019.

[15] S. Garbarino, et al., "Investigating hypotheses of neurodegeneration by learning dynamical systems of protein propagation in the brain," *NeuroImage*, vol. 235, 118024, 2021. doi: 10.1016/j.neuroimage.2021.118024.

[16] R. N. Gutenkunst, J. J. Waterfall, F. P. Casey, K. S. Brown, C. R. Myers, and J. P. Sethna, "Universally sloppy parameter sensitivities in systems biology models," *PLoS Computational Biology*, vol. 3, no. 10, p. e189, 2007, doi:10.1371/journal.pcbi.0030189.

[17] M. Hashemi, A. N. Vattikonda, V. Jirsa, and M. Woodman, "Simulation-based inference on virtual brain models of disorders," *Mach. Learn.: Sci. Technol.*, vol. 5, no. 3, 035019, 2024, doi:10.1088/2632-2153/ad6230.

[18] H.K. Kim, et al., "Temporal trajectory model for dopaminergic input to the striatal subregions in Parkinson's disease," *Parkinsonism & Related Disorders*, vol. 105, pp. 40–46, 2022. doi: 10.1016/j.parkreldis.2022.08.022.

[19] Kish, S.J., et al. (1988). Uneven pattern of dopamine loss in the striatum of patients with idiopathic Parkinson's disease. *New England Journal of Medicine*, 318(14), 876–880.

[20] C. Lee, J. Yoon, and M. van der Schaar, "Dynamic-DeepHit: A deep learning approach for dynamic survival analysis with competing risks based on longitudinal data," *IEEE Trans. Biomed. Eng.*, vol. 67, no. 1, pp. 122–133, Jan. 2020.

[21] M. Modrak, et al., "Simulation-based calibration checking for Bayesian computation: the choice of test quantities shapes sensitivity," *Bayesian Analysis*, vol. 1, no. 1, pp. 1–28, 2022. doi: 10.1214/22-BA1404.

[22] S. Pandya, et al., "Predictive model of spread of Parkinson's pathology using network diffusion," *NeuroImage*, vol. 192, pp. 178–194, 2019. doi: 10.1016/j.neuroimage.2019.02.058.

[23] J. Pasquini, et al., "Clinical implications of early caudate dysfunction in Parkinson's disease," *Journal of Neurology, Neurosurgery, and Psychiatry*, vol. 90, no. 6, pp. 681–687, 2019. doi: 10.1136/jnnp-2018-320157.

[24] J.R. Patterson, et al., "Time course and magnitude of alpha-synuclein inclusion formation and nigrostriatal degeneration in the rat model of synucleinopathy triggered by intrastriatal alpha-synuclein preformed fibrils," *Neurobiology of Disease*, vol. 121, pp. 1–12, 2019. doi: 10.1016/j.nbd.2018.09.001.

[25] F. Powell, M. Tosun, R. Sadeghi, P. Weiner, and A. Raj, "Preserved structural network organization mediates pathology spread in Alzheimer's disease despite loss of white matter tract integrity," *Journal of Alzheimer's Disease*, vol. 65, pp. 747–764, 2018. doi: 10.3233/JAD-180515.

[26] Parkinson Progression Marker Initiative (2011). The Parkinson Progression Marker Initiative (PPMI). *Progress in Neurobiology*, 95(4), 629–635.

[27] A. Raj, A. Kuceyeski, and M. Weiner, "A network diffusion model of disease progression in dementia," *Neuron*, vol. 73, no. 6, pp. 1204–1215, 2012.

[28] A. Raue, C. Kreutz, T. Maiwald, J. Bachmann, M. Schilling, U. Klingmüller, and J. Timmer, "Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood," *Bioinformatics*, vol. 25, no. 15, pp. 1923–1929, 2009, doi:10.1093/bioinformatics/btp358.

[29] J. Ren, L. Bhatt, G. Guo, and A. Raj, “Connectome-based biophysical models of pathological protein spreading in neurodegenerative diseases,” *PLoS Comput. Biol.*, vol. 21, no. 1, e1012743, 2025, doi:10.1371/journal.pcbi.1012743.

[30] A. Schafer, et al., “Bayesian physics-based modeling of tau propagation in Alzheimer’s disease,” *Frontiers in Physiology*, vol. 12, 702975, 2021. doi: 10.3389/fphys.2021.702975.

[31] M.J. Simpson, et al., “Making predictions using poorly identified mathematical models,” *Bulletin of Mathematical Biology*, vol. 86, no. 7, 83, 2024. doi: 10.1007/s11538-024-01294-0.

[32] C. Sung, et al., “Longitudinal decline of striatal subregional [18F]FP-CIT uptake in Parkinson’s disease,” *Nuclear Medicine and Molecular Imaging*, vol. 50, no. 4, pp. 314–320, 2016. doi: 10.1007/s13139-016-0436-4.

[33] S. Talts, et al., “Validating Bayesian inference algorithms with simulation-based calibration,” *arXiv preprint arXiv:1804.06788*, 2018.

[34] L. Tossici-Bolt *et al.*, “[^{123}I]FP-CIT ENC-DAT normal database: the impact of the reconstruction and quantification methods,” *EJNMMI Phys.*, vol. 4, Art. no. 8, 2017.

[35] A. F. Villaverde, A. Barreiro, and A. Papachristodoulou, “Structural identifiability of dynamic systems biology models,” *PLoS Computational Biology*, vol. 12, no. 10, p. e1005153, 2016, doi:10.1371/journal.pcbi.1005153.

[36] Vogel, J.W., et al. (2023). Connectome-based modelling of neurodegenerative diseases: towards precision medicine and mechanistic insight. *Nature Reviews Neuroscience*, 24(10), 620–639.

[37] S. Wang *et al.*, “Conditional neural ordinary differential equations for brain morphometry dynamics in Parkinson’s disease,” *arXiv*, 2025; 2410.23429.

[38] F.-G. Wieland, A. L. Hauber, M. Rosenblatt, C. Tönsing, and J. Timmer, “On structural and practical identifiability,” *Current Opinion in Systems Biology*, vol. 25, pp. 60–69, 2021.

[39] Xu, Y., et al. (2024). Kinetic analysis of alpha-synuclein aggregation. *Nature Communications*, 15, 7083.

[40] Zheng, Y.Q., et al. (2019). Local vulnerability and global connectivity jointly shape neurodegenerative disease propagation. *PLoS Biology*, 17(11), e3000357.